8. VERIFICATION OF NEWTON'S LAW OF COOLING USING CALORIMETER

AIM

To study the relationship between the temperature of a hot body and time by plotting a cooling curve.

APPARATUS REQUIRED

Copper calorimeter with stirrer, one holed rubber cork, thermometer, stop clock, heater / burner, water, clamp and stand

NEWTON'S LAW OF COOLING

Newton's law of cooling states that the rate of change of the temperature of an object is proportional to the difference between its own temperature and the ambient temperature. (i.e., the temperature of its surroundings)

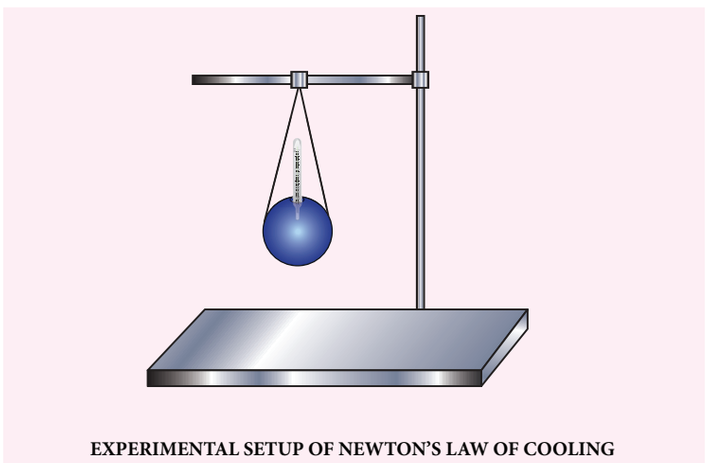
$$\frac{dT}{dt} \propto (T - T_0)$$

where $\frac{dT}{dt}$ ← Rate of change of temperature (°C)

T ← Temperature of water (°C)

T_0 ← Room Temperature (°C)

DIAGRAM



EXPERIMENTAL SETUP OF NEWTON'S LAW OF COOLING

PROCEDURE

- Note the room temperature as (T_0) using the thermometer.
- Hot water about 90°C is poured into the calorimeter.
- Close the calorimeter with one holed rubber cork.
- Insert the thermometer into calorimeter through the hole in rubber cork.
- Start the stop clock and observe the time for every one degree fall of temperature from 80°C.

UNIT-12(XI-Physics_Vol-2) PRACTICAL FIRST YEAR.indd 300 02-01-2023 15:28:05